SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 1

Fruit Battery

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Familiar objects make electricity

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

Session 1 — Build a Fruit Battery

Hook: Familiar objects make electricity.

Learning outcomes

By the end of this session, students should be able to:

- Build a simple electrochemical cell from fruit and metal electrodes
- Explain oxidation and reduction in a galvanic cell
- Identify anode and cathode and describe electron vs. ion flow
- Measure voltage with a multimeter
- Explain why cells in series add voltage
- Describe why batteries eventually die (internal resistance, reactant depletion)

Session plan (90 min)

Time	Activity	Notes
0–10 min	Hook: measure one lemon cell; ask whether a lemon can make electricity	See lecture.md — Opening hook
10–25 min	Concept: zinc oxidation, copper cathode, acid electrolyte, electron flow	See lecture.md
25–45 min	Build one cell; measure voltage	See experiment.md — Part 1
45–60 min	Series battery challenge	See experiment.md — Part 2
60–75 min	Power LED, buzzer, or clock; discuss V, I, and internal resistance	See experiment.md — Part 3
75–90 min	Side experiments: fruits, metals, pH, salt, parallel; wrap-up + preview Session 2	See experiment.md — Extensions

Key reactions

```
Anode:  Zn(s) → Zn²⁺(aq) + 2e⁻
Cathode: 2H⁺(aq) + 2e⁻ → H₂(g)
Overall: Zn(s) + 2H⁺(aq) → Zn²⁺(aq) + H₂(g)
```

Files in this folder

File	Purpose
lecture.md	~20 min concept block
experiment.md	Hands-on procedures and data
materials.md	Session checklist
preparation.md	Pre-class setup

Instructor reminders

- Roll lemons/limes first to break internal membranes
- Cells in series add voltage ~linearly; current stays limited
- Galvanized nail = steel + zinc coating (zinc is the active metal)
- pH near cathode may become less acidic (protons consumed)

Status

- [ ] Lecture outline finalized
- [ ] Experiment procedure tested
- [ ] Materials procured
- [ ] Session 2 $CuSO_4$ crystals on hand for next day

LECTURE TALKING POINTS

Session 1 — Lecture: Fruit Battery Concepts

Target duration: ~20 minutes (may occur before, during, or after the experiment)

> **Instructors:** for atomic structure, activity series, anode/cathode signs, and coaching notes, see instructor.md.

Opening hook (0–10 min)

Demo

1. Insert zinc and copper into one lemon (rolled first)
2. Connect multimeter; read voltage (~0.8–1.0 V typical)
3. Ask: "*Where did this voltage come from? Is the lemon a battery?*"

Guiding questions

- What is moving — electrons, ions, or both?
- Which electrode is positive on the meter? Which is negative?
- Would this work with other fruits? Other metals?

Core concepts (10–25 min)

1. Oxidation and reduction

Term	Meaning	In this cell
Oxidation	Loss of electrons	$Zn \rightarrow Zn^{2+} + 2e^-$
Reduction	Gain of electrons	$2H^+ + 2e^- \rightarrow H_2$
Redox	Coupled transfer of electrons	One cannot happen without the other

Mnemonic: OIL RIG — Oxidation Is Loss, Reduction Is Gain (of electrons)

2. Anode and cathode

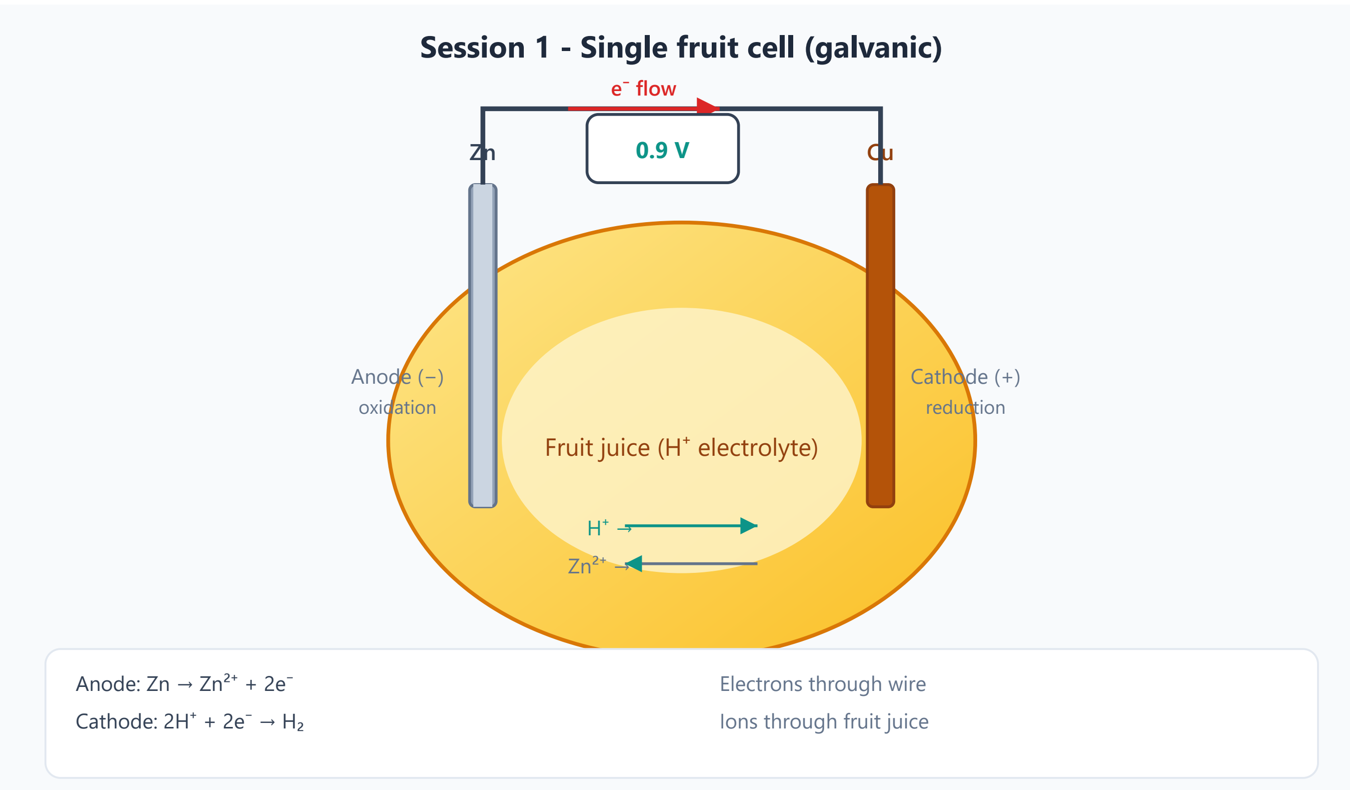
Electrode	Role	Sign in a galvanic cell	What happens here
Anode	Oxidation	Negative (−)	Zn loses electrons
Cathode	Reduction	Positive (+)	H^+ gains electrons

> **Common confusion:** In electrolysis (Session 2+), the anode is connected to the **positive** terminal of the power supply. In a **battery** (galvanic cell), the anode is **negative**. Always ask: *spontaneous cell or driven*

> *cell?*

3. Electron flow vs. ion flow

- **Electrons** travel through the **external wire** (metal path): anode → cathode
- **Ions** move through the **electrolyte** (fruit juice) to complete the circuit
- Without ion flow inside the fruit, the cell stops — charge builds up at the electrodes



See experiment.md for build steps and the series-cell schematic.

4. Voltage

- Voltage = electrical "push" from the chemical difference between the two metals
- Depends on: metal pair, electrolyte, temperature, electrode surface area
- **Not** the same as current or power

5. Cells in series

- Connecting cells **positive to negative** adds voltages
- 3 lemons × ~0.9 V ≈ 2.7 V — enough to try an LED (often needs ~2 V+)
- Current is still limited by the weakest cell and internal resistance

6. Why batteries die

- Reactants get used up (Zn dissolves, H^+ consumed)
- Internal resistance increases as surfaces change
- Side reactions and polarization reduce efficiency

Discussion prompts (use during lab)

- Why roll the fruit first?
- Why must the two electrodes **not touch** inside the fruit?
- What happens if you swap zinc and copper?
- Why might an LED flicker or not light even when V looks high enough?
- What is **internal resistance**? (Relate to why a lemon cell has low current)

Visual aids to prepare

- [ ] Whiteboard diagram: single cell with e^- and ion paths labeled
- [ ] Table: anode vs. cathode in galvanic vs. electrolytic cells
- [ ] Photo or live demo: multimeter reading on one cell vs. three in series
- [ ] Optional: pH strips showing less acidic region near cathode

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Vocabulary checklist

Ensure students can define before leaving:

- [ ] Oxidation / reduction
- [ ] Anode / cathode (in a **battery**)
- [ ] Electrolyte
- [ ] Voltage
- [ ] Series connection
- [ ] Internal resistance

Bridge to Session 2

End concept block with a preview:

"Today we used chemistry to make electricity. Next session we'll use electricity to force chemistry — moving copper atoms onto a nail from a copper sulfate solution."

INSTRUCTOR DEEP-DIVE NOTES

Session 1 — Instructor Notes: Fruit Battery Deep Dive

Audience: instructors and TAs only. Students see the shorter lecture.md and experiment.md. Use this page to answer “why?” with confidence.

Why this session exists

Students already know “batteries make electricity.” The fruit cell makes the chemistry visible: two different metals, an acidic juice, a measurable voltage, and (with luck) a tiny load. Your job is to connect that demo to **electrons leaving metal atoms**, **ions moving in solution**, and the idea that **oxidation and reduction are always paired**.

Atomic and electronic foundations

Atoms, electrons, and valence shells (what students need)

Keep this brief but precise if students lack chemistry background:

1. Atoms have a **nucleus** (protons + neutrons) and **electrons** outside.
2. Chemistry of batteries is almost entirely about the **outermost (valence) electrons**.
3. Metals tend to **lose** valence electrons → become **positive ions** (cations).
4. In solution, those electrons do not vanish — they travel through a wire and are gained by something else (reduction).

Zinc and copper electron structure (instructor level)

Metal	Atomic number	Valence electrons (simplified)	Typical ion	What that means here
Zn	30	2 ($4s^2$)	Zn^{2+}	Readily loses $2e^-$ → dissolves as Zn^{2+}
Cu	29	1 ($4s^1$) in free atom; chemistry often treated as Cu^{2+}/Cu	Cu^{2+} or Cu^+	Less eager than Zn to dissolve in this cell

Teaching tip: You do **not** need full electron configurations with students. Say: *“Zinc holds its outer electrons less tightly than copper in this setup, so zinc is the metal that oxidizes.”*

What “oxidation” means at the atom scale

At the zinc surface:

```
Zn(s) → Zn²⁺(aq) + 2e⁻
```

- The solid zinc atom leaves the metal lattice as Zn^{2+} into the juice.
- The two electrons stay in the metal and can travel through the external wire.

- That is why the zinc electrode is the **anode** (oxidation site) and, in a galvanic cell, the **negative** terminal (source of electrons into the external circuit).

At the copper surface (simplified classroom story for acidic fruit juice):

```
2H⁺(aq) + 2e⁻ → H₂(g)
```

- Protons from the acidic juice gain electrons and become hydrogen gas.
- Copper mainly provides a surface for reduction; it is **not** the main reactant being consumed (though real cells can have side reactions).

Nuance for you (not always for students): In some fruit cells, dissolved oxygen or other species also accept electrons. The $H^+ \rightarrow H_2$ story is the clean teaching model; bubbles may be subtle.

Periodic trends and the electrochemical series

Activity / reactivity series (classroom version)

More active metals oxidize **preferentially**:

```
Mg > Al > Zn > Fe > Sn > Pb > (H) > Cu > Ag > Au
```

How to use this in Session 1:

- Zn is above Cu → Zn oxidizes; Cu is the better cathode surface.
- Swap electrodes in discussion: if both metals are the same, $\Delta E \approx 0$.
- If students try Al or Mg: expect different voltages and often messy oxide layers (Al_2O_3 film can block current).

Standard reduction potentials (instructor reference)

Approximate aqueous standard potentials (vs SHE, acidic):

Half-reaction	E° (V)
$Zn^{2+} + 2e^- \rightarrow Zn$	−0.76
$2H^+ + 2e^- \rightarrow H_2$	0.00
$Cu^{2+} + 2e^- \rightarrow Cu$	+0.34

Cell potential (ideal Zn/H^+ picture):

```
E°_cell ≈ E°_cathode − E°_anode(as oxidation)
       ≈ 0.00 − (−0.76) = +0.76 V
```

Students measure ~0.8–1.0 V — close enough for a lemon. Real fruit cells are **not** standard conditions (unknown $[H^+]$, mixed ions, internal resistance), so do not oversell exact E° matching.

Key instructor point: Voltage measures a **difference in chemical preference** for holding electrons, not "how much electricity is stored."

Lecture concepts — expanded

Galvanic cell definition

A **galvanic (voltaic) cell** converts chemical energy into electrical energy spontaneously:

- Oxidation at anode, reduction at cathode
- Electrons flow **anode → cathode** in the external wire
- Ions migrate in the electrolyte to keep charge balanced

Without ion motion, the circuit is incomplete: electrons pile up on one side, charge separation stops the reaction.

Anode / cathode sign convention (the #1 confusion)

Cell type	Anode process	Anode sign	Cathode process	Cathode sign
Galvanic (Session 1)	Oxidation	**−**	Reduction	**+**
Electrolytic (Session 2+)	Oxidation	**+** (connected to battery +)	Reduction	**−**

Memory aid for instructors: *Anode = oxidation always. Sign depends on whether the cell is spontaneous or driven.*

Voltage vs current vs power

Quantity	What it is	Fruit cell reality
Voltage (V)	Electrical potential difference	Often ~0.8–1.0 V per cell
Current (I)	Charge flow rate	Usually tiny (μA–mA)
Power (P = VI)	Energy per time	Too small for most loads unless many cells / good contacts

Students often think "1 V is a lot." Emphasize: a AA battery is ~1.5 V **and** can supply much higher current. Lemons fail LEDs more from **current / internal resistance** than from voltage alone.

Internal resistance

Model a real cell as an ideal voltage source in series with resistance (R_\text{int}):

```
V_load = E − I·R_int
```

When a load draws current, measured voltage **drops**. Rolling fruit, clean metal, larger electrode area, and better juice contact all lower (R_\text{int}).

Series vs parallel

- **Series:** voltages add; same current through each cell. Needed for LED (~2 V).

- **Parallel:** voltage stays ~one cell; current capacity increases (in ideal theory). Fruit cells rarely reward parallel setups because contacts and mismatches dominate.

Experiment notes — what to watch for

Build quality that actually matters

1. **Roll the fruit** — ruptures juice sacs; improves ion pathways.
2. **Gap between electrodes** — touching shorts the cell (0 V externally).
3. **Clean metal** — oxide/dirt raises resistance.
4. **Clip polarity** — decide a convention (e.g. red → Cu) and stick to it so series wiring is consistent.

Typical numbers (pilot these yourself)

Setup	Expected
Lemon + Zn + Cu	~0.8–1.0 V open circuit
Potato / apple	Often slightly lower or similar
3 cells in series	~2.4–3.0 V open circuit
Voltage under LED load	Often collapses if current demand is high

Why the LED may fail even at “enough volts”

- LED needs both **forward voltage** (~1.8–2.2 V for red) **and** a few mA.
- Fruit cell can show 2.7 V open-circuit but sag under load.
- Wrong LED polarity → no light.
- Prefer **red diffused LEDs**; blue/white need higher voltage.

Extension: metal comparisons (how to narrate results)

If students replace Zn with Fe or Al:

- Fe vs Cu: smaller voltage than Zn vs Cu (Fe less active than Zn).
- Al vs Cu: theoretically high activity, but Al_2O_3 film often suppresses performance — great teachable moment about **surface chemistry**, not just the activity table.

Bridge to Session 2 (no overnight copper prep)

Session 2 uses **dissolved copper sulfate ($CuSO_4$)** mixed day-of (or shortly before). Do not start vinegar/salt copper jars at the end of Session 1. Preview only: tomorrow electricity will force Cu^{2+} ions onto a nail.

Misconceptions to preempt

Misconception	Better framing
“The lemon is the battery.”	The lemon is the **electrolyte**; the metals drive the chemistry.

Misconception	Better framing
"Electricity is stored in the lemon."	Energy comes from the **chemical reaction**, mainly Zn oxidation.
"Electrons travel through the juice."	Electrons travel in the **wire**; **ions** travel in the juice.
"Anode is always +."	Only true for driven electrolytic cells; here anode is −.
"Higher voltage always means brighter LED."	Current and internal resistance matter as much as voltage.

Board plan (20 min concept block)

1. Draw lemon + Zn + Cu; label anode/cathode.
2. Write Zn oxidation and H^+ reduction half-reactions.
3. Draw external e^- path and internal ion path (two different arrows).
4. Show one cell vs three in series.
5. Preview Session 2: *tomorrow we reverse the idea — electricity moves metal atoms.*

Optional enrichment (if a strong student asks)

- Nernst equation qualitatively: concentration and pH shift voltage.
- Salt bridges vs fruit electrolyte (why Daniell cells use two compartments).
- Why commercial batteries use solid electrolytes / pastes (leakage, shelf life, current density).

PREPARATION REMINDERS

Session 1 — Preparation

Before class (instructor)

- [ ] Procure fresh fruit (firm, not dried out)
- [ ] Test one cell: expected ~0.8–1.0 V for lemon + Zn + Cu
- [ ] Verify multimeters have fresh batteries and work on DC voltage
- [ ] Pre-cut copper strips if students struggle with stiff wire
- [ ] Prepare waste collection for fruit and electrodes

Day-of setup (15 min before)

- [ ] One station per group: fruit, electrodes, wires, multimeter, load device
- [ ] Optional extension materials at a side table

After class

- [ ] Dispose of fruit waste appropriately
- [ ] Rinse and dry electrodes if reusing
- [ ] Confirm Session 2 $CuSO_4 \cdot 5H_2O$ is on site for tomorrow's plating bath

Open questions / notes

- Typical voltage achieved with our materials:
- Best fruit type in pilot:
- LED lighting requires ___ cells in series: